AsmFEA - Getting started

Stress analysis of a whole SOLIDWORKS assembly, from a task pane, in a few clicks.

1 Find the pane | 2 Prepare | 3 Loads & fixtures | 4 Connections | 5 Run | 6 Results | 7 Fatigue

1 Find AsmFEA in SOLIDWORKS

1. Start SOLIDWORKS (2018 or later; one of the versions you ticked in the installer). AsmFEA loads with it.
2. On the right of the window, the task pane has a new tab with the coloured hexagon icon (the one at the top of this page): click it to open AsmFEA.
3. If it is missing: **Tools > Add-ins**, tick **AsmFEA** in both columns, then OK.

2 Prepare the assembly and start a study

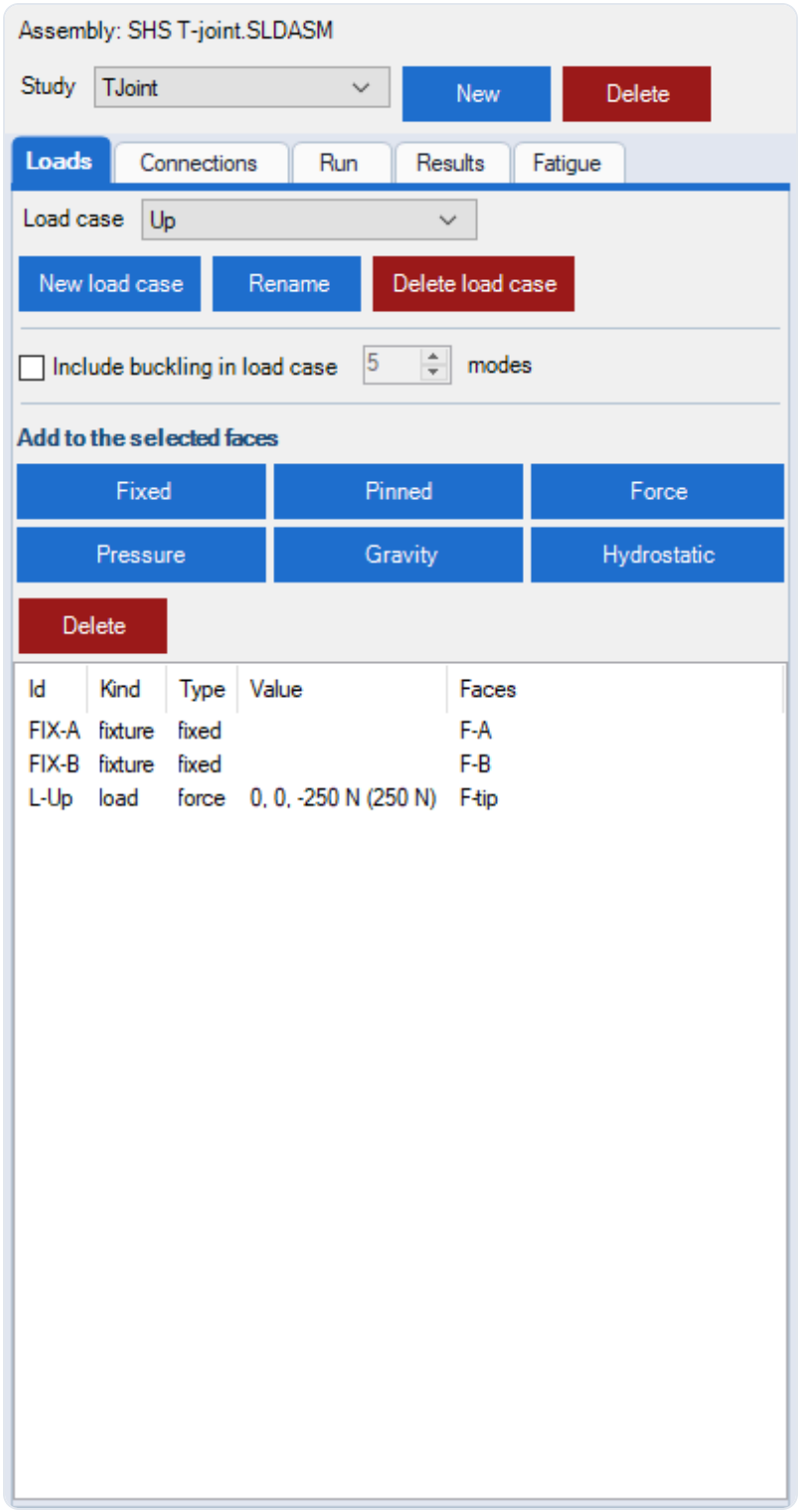
- Open the assembly and give every part a material (right-click the part > **Material**). A single part works too: its bodies are treated as the parts, and bodies that touch are joined.
- Click `New` in the pane and name the study. It is saved inside the assembly (or part). Every setting is kept as you make it.
- The pane has five tabs, used in order: **Loads**, **Connections**, **Run**, **Results** and **Fatigue**. The main button on each tab is the dark blue one; red buttons remove something.

> Parts that touch are joined automatically. Mates tell AsmFEA where parts can slide or turn; weld beads are used too, and a sub-assembly is treated as one rigid piece.

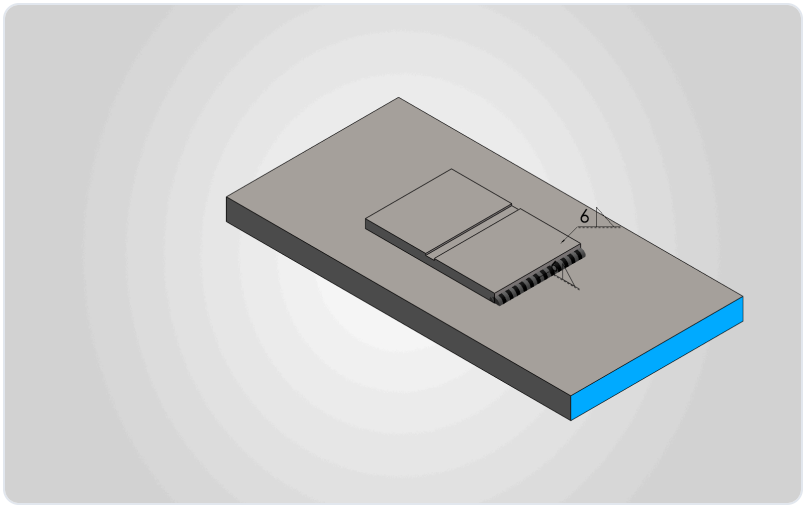
3 Add fixtures and loads

1. Select one or more faces on the model.
2. On the **Loads** tab, under *Add to the selected faces*, click `Fixed` (held still) or `Pinned` (can rotate).
3. Loads belong to a **load case** (a set of loads solved together). Pick one at the top of the tab, or click `New load case`; every case is solved in the same run. Fixtures apply to all cases. Tick **Include buckling in load case** to check a case for buckling too (runs take longer).
4. Select faces and click `Force` or `Pressure`, or click `Gravity` for self-weight. Give the force as X, Y and Z components; the resultant shows as you type.
5. For a force acting away from the faces (a lever arm, a torque), also give how far from the faces' centre it acts, in X, Y and Z (leave 0, 0, 0 for a force on the faces).
6. For fluid in a tank, select the wetted faces and click `Hydrostatic`. Give the fluid's density (water 1000 kg/m^3), the absolute pressure on its top (1 atmosphere, 0.101325 MPa, for an open tank; the air outside balances it), and where the top of the fluid is, in X, Y and Z from the faces' centre; a ball and triad marks it on the model. The pressure rises with depth below it; faces above it are not loaded.

Click a row in the list to highlight its faces on the model; a load also shows an arrow pointing at where it acts. Double-click a row to change it.



The Loads tab

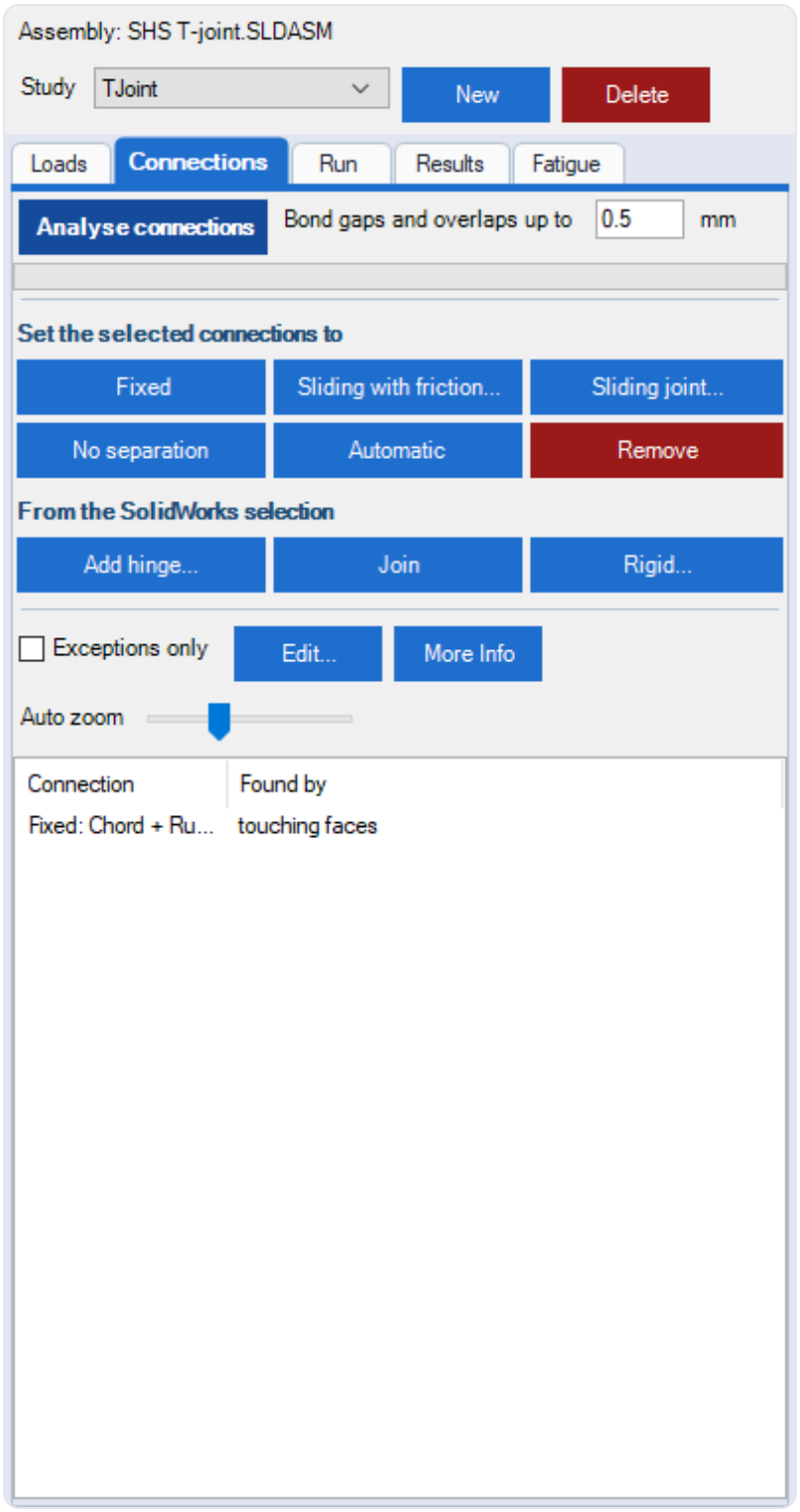


The fixed face highlighted in blue. This assembly is a plate on a base, held by a 6 mm weld bead along one edge.

4 Check how the parts connect

- Click `Analyse connections`: the list shows every place where parts meet and how they will be joined.
- Click a connection to see it on the model: red balls mark the faces that join, and the view zooms to it. The **Auto zoom** slider sets how close: far left shows the whole model, far right the connection's faces.
- To change one, select it and click `Edit...`, or use *Set the selected connections to*: `Fixed` (bonded), `Sliding with friction...`, `Sliding joint...` or `No separation` (sliding, never parting). `Remove` disconnects the parts there; `Automatic` undoes your choice.
- *From the SolidWorks selection*: `Add hinge...` makes a pin joint on a hole or pin face, `Join` connects two parts that do not touch, and `Rigid...` treats parts whose shape is only approximate as rigid (they carry load and weight, but are not meshed).
- Weld beads appear as their own rows.
- On a large model (over 50 parts) every part is first bonded to the parts it touches, and mates are applied where they free a motion that matters; small parts hanging off the structure are left out and listed in the report. *Advanced* > *Connections* on the Run tab chooses the method yourself.
- If parts are still floating, a window lists them with the gap to their nearest neighbour: raise the bond tolerance to join them, or leave them out of the analysis (they stay in SolidWorks).

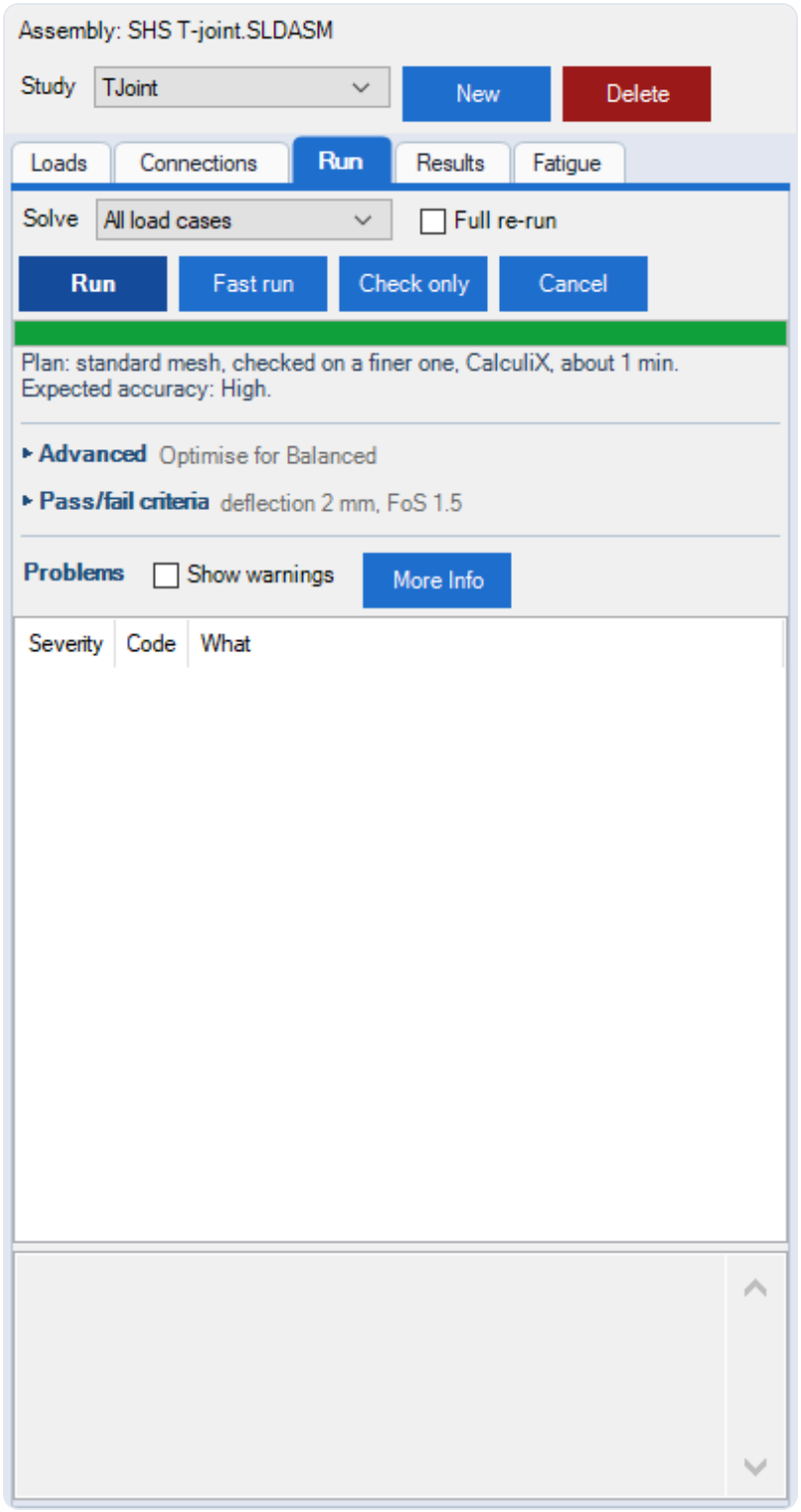
> Hover over any button for a tip. Double-click a connection or a warning, or press `?`, for a full explanation.



The Connections tab

5 Run

- On the **Run** tab click `Run`. The bar and the line below it show what is happening; a first run takes from under a minute to a few minutes. The bar is green while the results match the study as it is, and red when something has changed since.
- **Advanced** (click to open) sets what to **Optimise for**: *Speed* (a coarse mesh, one solve: quick, for comparing designs), *Balanced* (for most work) or *Accuracy* (a finer mesh for local stresses, and parts whose results have not settled solved again finer). AsmFEA decides the rest from this and your computer's free memory and cores, always leaving room for SolidWorks and other programs; the plan shows under the buttons. You can also set the mesh and solver by hand there, and *Fast load-size changes*: each load solved once, so changing only load sizes afterwards needs no new solve.
- **Pass/fail criteria** (click to open) holds the limits the checks and reports use: maximum deflection and minimum factors of safety. Changing a limit re-checks the last results without solving again.
- `Check only` just checks the set-up, quickly, without solving.
- `Fast run` is for trying design changes: a coarser model that shows where the problem is and whether a change helped, in a fraction of the time. Its results are marked **APPROXIMATE** and it writes no report: do a full `Run` at the end, before you rely on the numbers.
- A re-run redoes only what your changes need: with nothing changed it finishes in seconds, and a new pass/fail limit skips the solve. Tick *Full re-run* to redo everything.
- A model too large for CalculiX to fit in your computer's memory is no longer refused: AsmFEA solves it with its own hybrid solver, part by part, using the memory it has free. A note on the Run tab and in the report says when that happened.
- A run is never stopped for memory. If other programs take memory while it runs, a box says the solve is continuing more slowly; closing programs you don't need speeds it up again.
- The results say how far to trust them: a coloured line on the **Results** tab, and the reports, grade them *High*, *Good* or *Indicative* with the likely error on displacements and stresses, from how the model was solved. Click the line to see why.
- Anything that needs attention is listed: double-click it to see what it means and what to do.



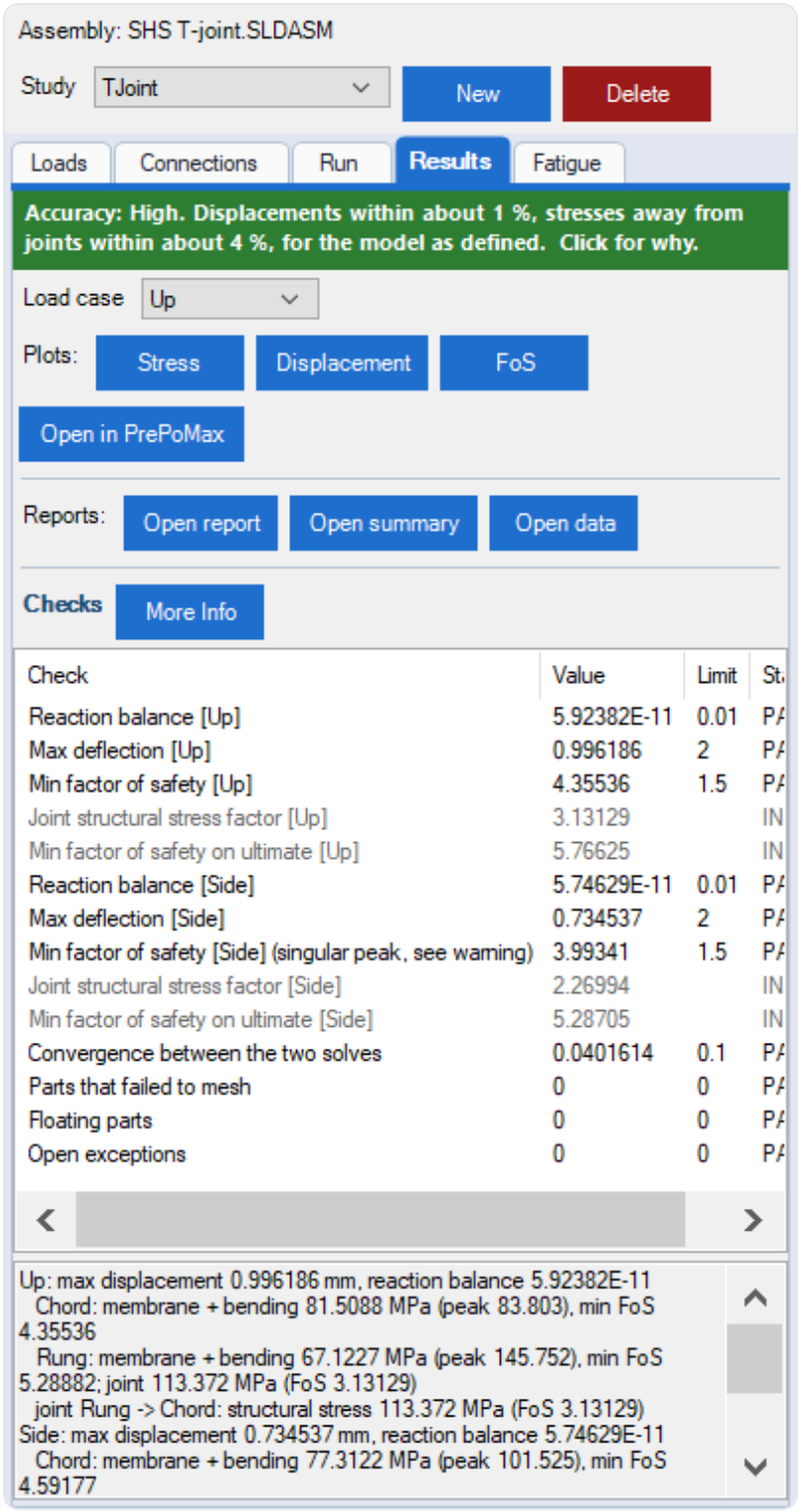
The Run tab: the plan, then Advanced and Pass/fail criteria, closed

6 Look at the results

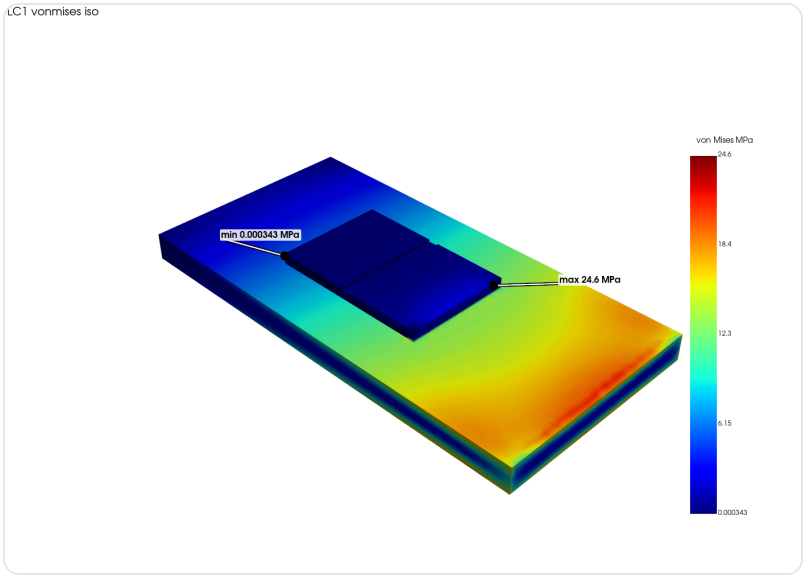
- The **Results** tab shows the pass/fail checks and the key numbers for each load case. Stress is judged as an engineer would: through the wall (membrane plus bending), round

joints as well, and not on the spikes at sharp edges. The factor of safety is given on yield and, for information, on ultimate strength.

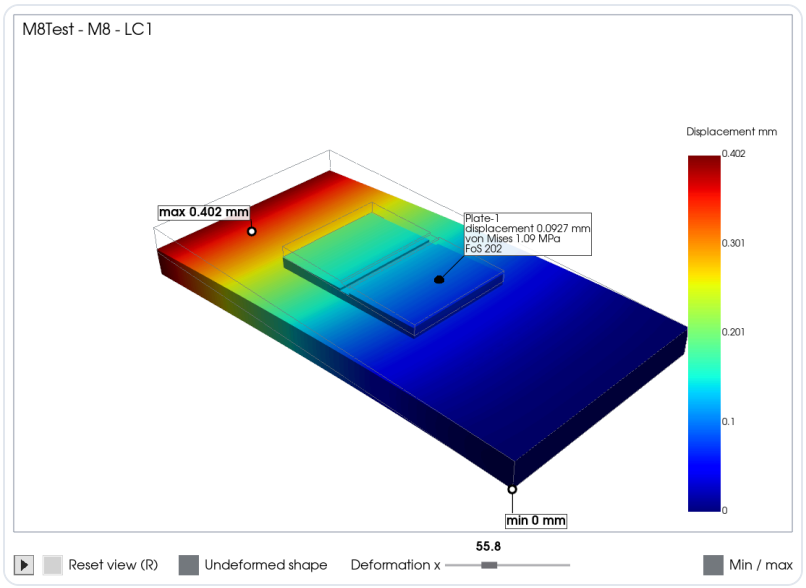
- `Stress`, `Displacement` and `FoS` open a 3-D window (their tooltips repeat what follows):
  - left drag to rotate, wheel to zoom, right drag to pan, as in SOLIDWORKS; **Reset view** (or R) goes back;
  - **click the model** to see the part and its stress, displacement and FoS at that point; click the background to clear it;
  - ▶ **play** ramps the load up and down so you can watch the stress build; click it again to stop at the full load;
  - **Deformation x** exaggerates the deformed shape, on every plot (0 shows it undeformed), and **Undeformed shape** adds a grey outline of the original;
  - **Min / max** shows the highest and lowest values; drag their labels anywhere;
  - **right-click a part** to see it as an outline and look at the parts behind it; right-click it again, or the background, to bring it back;
  - local peaks beyond the assessed stress show **magenta**; grey parts (such as bushes) are not assessed: check their bearing pressure by hand.
- `Open report` gives the Word report, `Open summary` a half-page summary, and `Open data` the data behind them (the model, materials, connections, forces and detailed results). They need a full run (they are greyed out after a fast run).
- The pictures follow the way you look at the model in SOLIDWORKS: turn it, then open the results again.


Assembly: SHS T-joint.SLDASM
Study
TJoint
New
Delete
Loads
Connections
Run
Results
Fatigue
Accuracy: High. Displacements within about 1 %, stresses away from joints within about 4 %, for the model as defined. Click for why.
Load case
Up
Plots:
Stress
Displacement
FoS
Open in PrePoMax
Reports:
Open report
Open summary
Open data
Checks
More Info
Check
Value
Limit
Reaction balance [Up] 5.92382E-11 0.01
Max deflection [Up] 0.996186 2
Min factor of safety [Up] 4.35536 1.5
Joint structural stress factor [Up] 3.13129
Min factor of safety on ultimate [Up] 5.76625
Reaction balance [Side] 5.74629E-11 0.01
Max deflection [Side] 0.734537 2
Min factor of safety [Side] (singular peak, see warning) 3.99341 1.5
Joint structural stress factor [Side] 2.26994
Min factor of safety on ultimate [Side] 5.28705
Convergence between the two solves 0.0401614 0.1
Parts that failed to mesh 0 0
Floating parts 0 0
Open exceptions 0 0
Up: max displacement 0.996186 mm, reaction balance 5.92382E-11
Chord: membrane + bending 81.5088 MPa (peak 83.803), min FoS 4.35536
Rung: membrane + bending 67.1227 MPa (peak 145.752), min FoS 5.28882; joint 113.372 MPa (FoS 3.13129)
joint Rung -> Chord: structural stress 113.372 MPa (FoS 3.13129)
Side: max displacement 0.734537 mm, reaction balance 5.74629E-11
Chord: membrane + bending 77.3122 MPa (peak 101.525), min FoS 4.59177

Results



Stress (von Mises), with the highest and lowest values marked

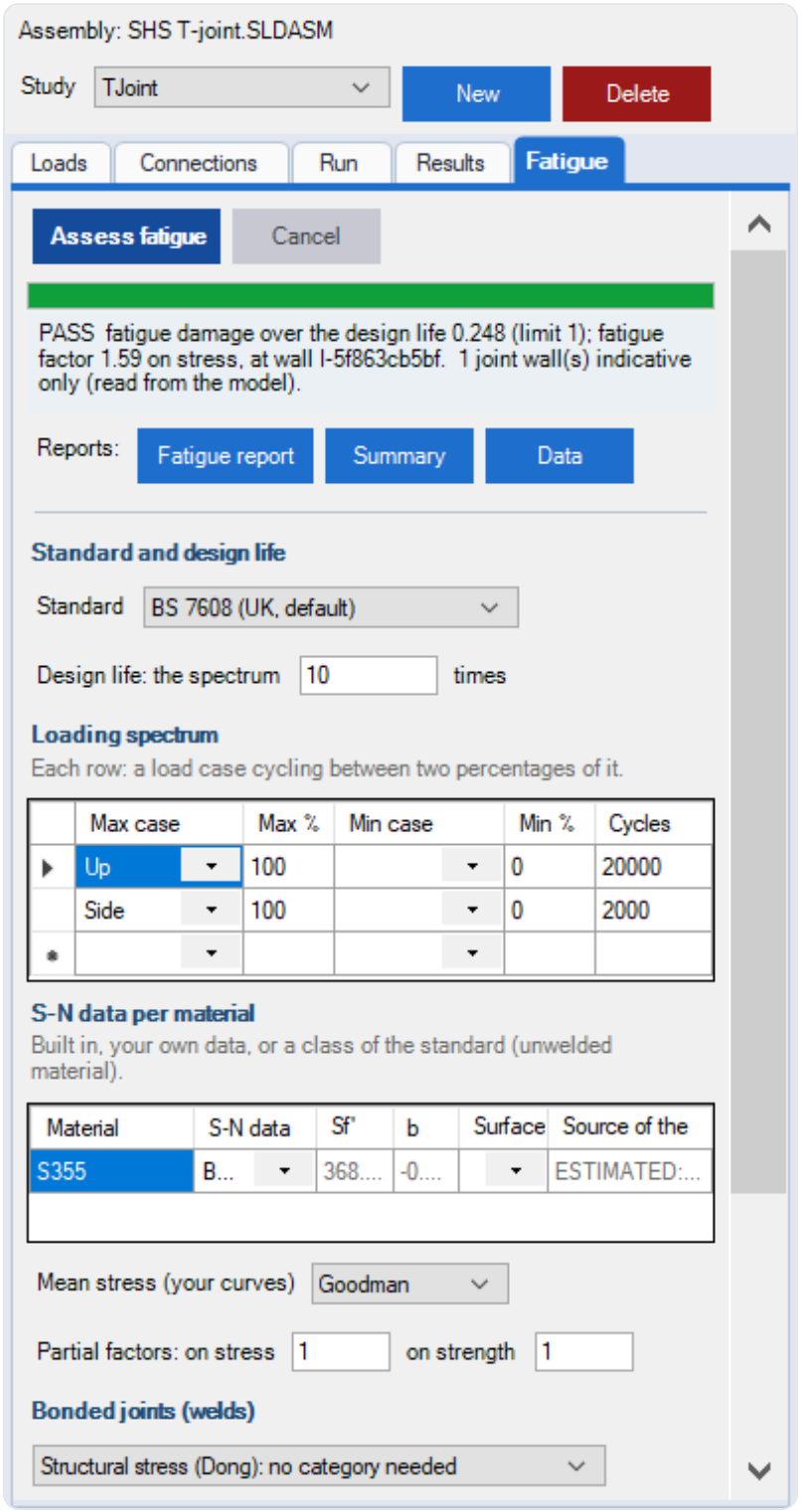


The 3-D window: displacement with the undeformed outline, a clicked point and the min / max labels

7 Fatigue

Fatigue is cracking under repeated loading, at stresses well below yield. It matters for anything loaded many times: lifting, travelling, vibrating or rotating parts, and even a rig that only has to last a few hours can see thousands of cycles. The **Fatigue** tab works it out from the static results, with no new solve.

1. Choose the **standard** (BS 7608 by default) and how many times the spectrum repeats in the design life.
2. Fill in the **loading spectrum**: each row is a load case cycling between two percentages of it (100 % to 0 %: the load put on and taken off) a number of times. Use one load case at both ends, unless the load moves between two positions.
3. The **S-N data** per material is built in: the SolidWorks material's own curve if it has one, otherwise published data, or an estimate from the tensile strength (marked ESTIMATED). Give your own if you have it.
4. Click `Assess fatigue`. The result shows the damage over the design life (1 means a crack is expected) and the fatigue factor: how much every stress could rise before that.
5. Bonded joints (welds) are listed weakest first, with the factors of the weld and of the wall under it: click one to see it on the model.
6. `Fatigue report`, `Summary` and `Data` open its own documents. Hover over any setting for an explanation.



The Fatigue tab after an assessment

8 Optional: PrePoMax, for a closer look

PrePoMax is a free program for exploring results in detail: section cuts, values at any point, animations. AsmFEA does not need it, but `Open in PrePoMax` on the Results tab opens the

last run in it.

1. Download it from prepomax.fs.um.si/downloads (version 2.6.0 is the one AsmFEA was tested with).
2. It is a zip file: unzip it anywhere, for example `C:\PrePoMax`. There is nothing to install.
3. In AsmFEA click Open in PrePoMax and choose **Find PrePoMax.exe...** the first time; AsmFEA remembers it.

✓ How accurate is AsmFEA?

The verification report (also in the Start menu, and as a PDF in the AsmFEA folder) compares AsmFEA with classical hand calculations on four test cases (a cantilever, a beam under its own weight, two members joined end to end, and a buckling column) and checks the CalculiX solver against its own test examples and two NAFEMS benchmarks. Every comparison came within its acceptance criterion, the largest difference being 0.6 %.

Satisfy yourself of its accuracy for your own work by running your own tests against hand calculations or other trusted results before relying on it.

9 Keep your work

- The study is saved **inside the assembly**: save the assembly to keep it.
- Results and reports (static and fatigue) are in a folder next to the assembly called `<assembly name>_FEA`.
- Units are millimetres, newtons and MPa throughout.

> Always check its results with engineering judgement and a hand calculation before relying on them for anything that matters.

AsmFEA uses CalculiX (solver) and Gmsh (mesher), free software downloaded by the installer; see the licenses folder.